

Credit Card Segmentation Analysis Report

Problem:

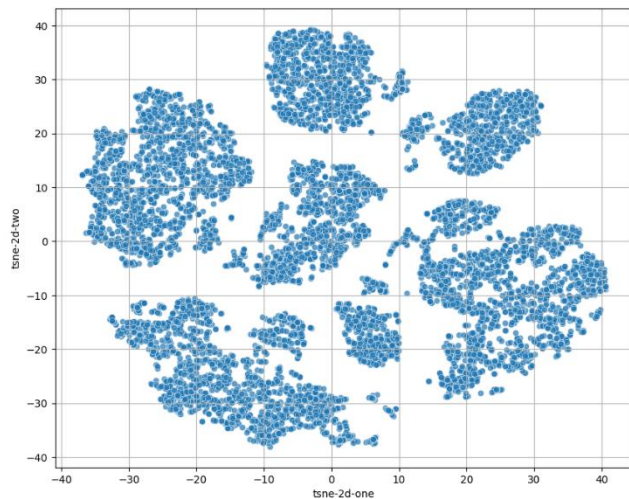
To segment the credit card customer base into distinct behavioral groups to enable targeted marketing and risk management.

How numbers of clusters were determined?

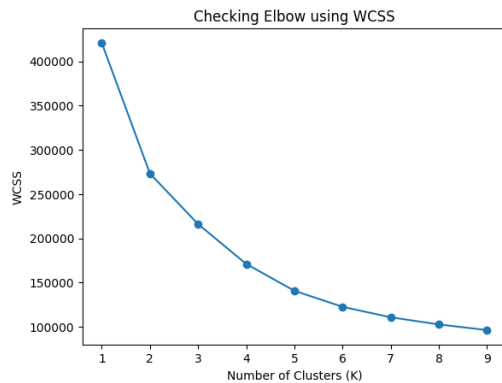
We determined number of clusters using two methods

T-SNE:

We visualized data in 2D dimension using TSNE and used our intuition to think of dense clusters here 5 to 7 would be a good number



WCSS (Elbow method)

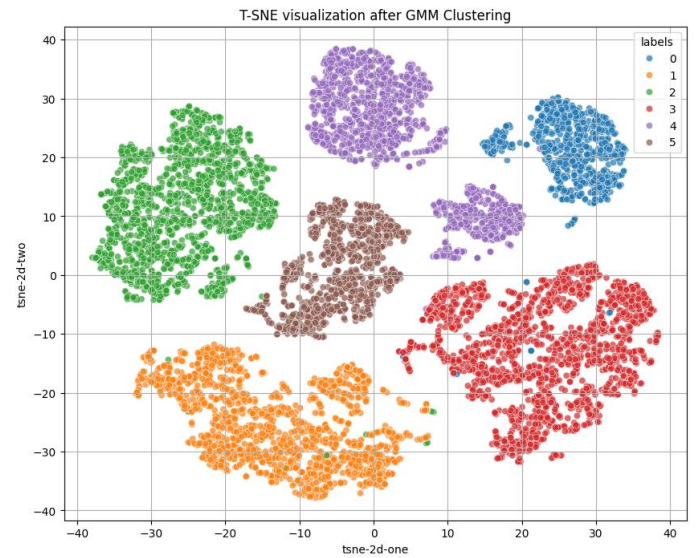
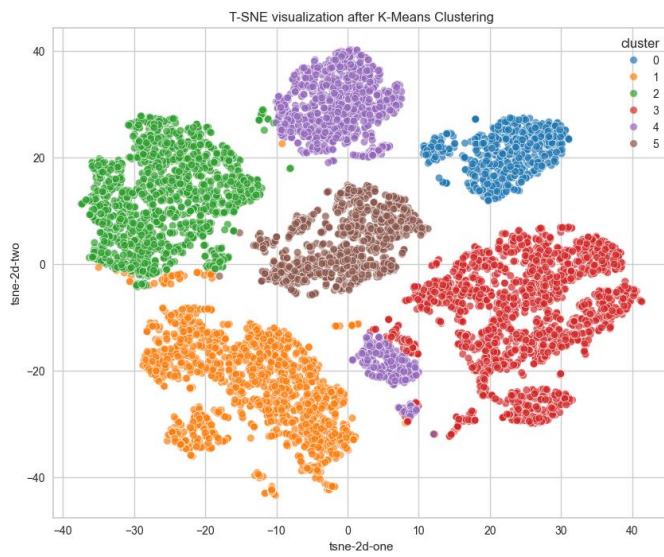


We have finally decided to go with 6 clusters

Models Used:

- K-means
- GMMs

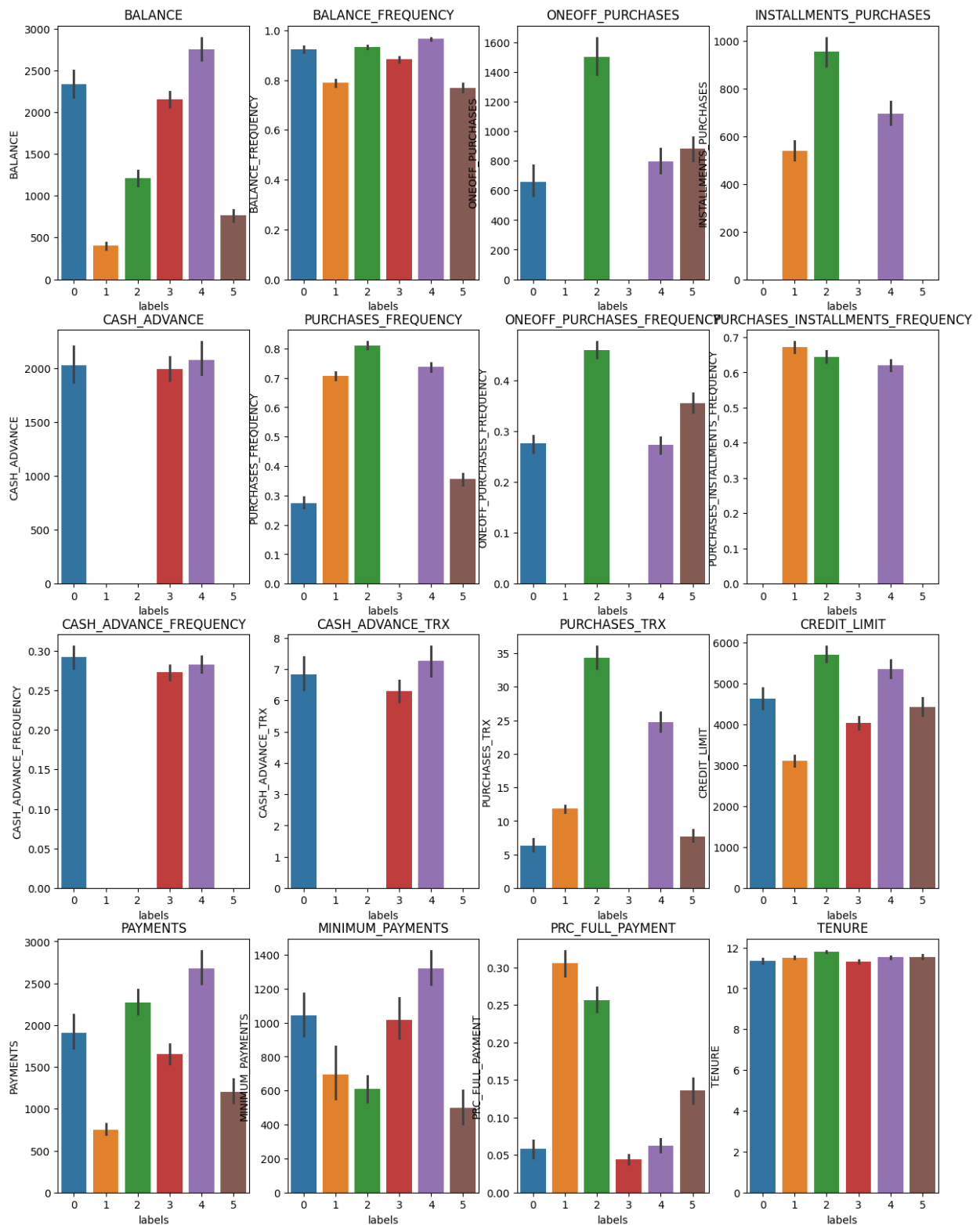
The following is visualizations of clusters made using T-sne



We can see that we get more separated clusters in GMM

Analysis of each segment:

Barplots of mean of each segment at each given feature:



Radarplots of most important statistics for each cluster:



Insights about clusters:

Cluster 0: Cash Advance Dependents

- Insights: This group has high **BALANCE** and **BALANCE_FREQUENCY**. They rarely use the card for retail purchases (**PURCHASES_TRX**, **ONEOFF_PURCHASES**, and **INSTALLMENTS_PURCHASES** are near zero). Instead, they heavily rely on cash (**CASH_ADVANCE**, **CASH_ADVANCE_TRX**). Their **PRC_FULL_PAYMENT** is extremely low, meaning they revolve their debt.
- Persona**: Customers using the credit card primarily as a high-interest loan or emergency cash fund.

Cluster 1: Cautious Installment Spenders

- **Insights:** This cluster maintains the lowest **BALANCE** and **CREDIT_LIMIT**. They avoid cash advances entirely. Their purchasing behavior is dominated by installments (**PURCHASES_INSTALLMENTS_FREQUENCY** is high). Notably, they have the highest **PRC_FULL_PAYMENT** rate.
- **Persona:** Financially conservative users who use the card to break down specific purchases into manageable payments and consistently pay off their bills.

Cluster 2: Premium Transactors

- **Insights:** These are your heaviest spenders for retail (**ONEOFF_PURCHASES**, **PURCHASES_TRX**), but they have very low **CASH_ADVANCE** activity. They have high **CREDIT_LIMITs** and high **PAYMENTS**. They also have a very high **PRC_FULL_PAYMENT** percentage.
- **Persona:** High-value, low-risk customers who use the card for daily high-volume spending and pay off their balances frequently to avoid interest.

Cluster 3: Struggling / High-Risk Revolvers

- **Insights:** Very similar to Cluster 0 (will have similar business plan), with high **CASH_ADVANCE** usage and almost zero retail purchases. However, they have a lower **CREDIT_LIMIT** and lower **BALANCE** than Cluster 0. They also struggle to pay the card off (**PRC_FULL_PAYMENT** is nearly zero).
- **Persona:** A higher-risk subgroup of cash-dependent users who likely have lower credit scores and are maxing out lower-limit cards.

Cluster 4: Mega Users (Debt Accumulators)

- **Insights:** This group does everything. They have the highest **BALANCE**, high **ONEOFF_PURCHASES**, high **INSTALLMENTS_PURCHASES**, and high **CASH_ADVANCE**. They possess the highest **CREDIT_LIMIT** but have a very low **PRC_FULL_PAYMENT** rate, meaning they are carrying massive amounts of debt across all transaction types.
- **Persona:** Highly lucrative but potentially high-risk customers who max out the card's utility across purchases and cash advances while carrying over balances.

Cluster 5: Occasional One-Off Spenders

- **Insights:** This group has a low **BALANCE** and makes very few transactions (**PURCHASES_TRX** is low). However, when they do buy, it is almost exclusively one-off purchases (**ONEOFF_PURCHASES_FREQUENCY** is noticeable, while **installment frequency** is zero). They have moderate credit limits but low overall engagement.
- **Persona:** Dormant or infrequent users who keep the card in their wallet for occasional, specific large purchases.

Business strategies for each cluster:

For Clusters 0 & 3 (Cash Advance / High-Risk):

- **Strategy:** Risk mitigation and debt restructuring.
- **Actions:** Do not increase their credit limits. Offer them personal loans or balance consolidation programs at slightly lower interest rates to convert high-risk revolving credit into structured installment debt. Block promotional offers that encourage further spending.

For Cluster 1 (Cautious Installment Spenders):

- **Strategy:** Increase wallet share and gradual limit expansion.
- **Actions:** Offer "Buy Now, Pay Later" partnerships with specific retailers (e.g., electronics, furniture). Since they are reliable payers, gradually increase their credit limits. Offer point multipliers for everyday non-installment purchases (like groceries) to encourage broader card utility.

For Cluster 2 (Premium Transactors):

- **Strategy:** Retention and lifestyle rewards.
- **Actions:** These users generate revenue through interchange fees, not interest. Target them with premium rewards (airport lounge access, travel points, luxury concierge services). Ensure their credit limit is actively managed upward to prevent them from hitting utilization caps and switching to a competitor's card.

For Cluster 4 (Mega Users):

- **Strategy:** Yield maximization with strict monitoring.
- **Actions:** They are highly profitable due to interest payments, but their high balances pose a default risk. Monitor their credit scores externally. Offer them targeted promotions to shift their cash-advance behavior (which usually carries higher default risk) toward high-value retail purchases.

For Cluster 5 (Occasional One-Off Spenders):

- **Strategy:** Activation and habit-building.
- **Actions:** Run targeted campaigns designed to build daily habits, such as cashback on gas, dining, or subscription services (Netflix, Spotify). Offer limited-time "spend \$X, get \$Y back" promotions to reactivate their daily usage.